IAU

PYTHON-BASED SENTIMENT ANALYZER

Analyzing Amazon Product Reviews & Multilingual Support

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Introduction

Sentiment analysis, also known as opinion mining, is a pivotal technique in Natural Language Processing (NLP) that identifies the emotional intent behind a body of text. In the contemporary e-commerce landscape, customer reviews serve as a primary source of data for businesses. However, the exponential growth of unstructured text data makes manual analysis inefficient and prone to human error.

This report walks through how we built an AI tool to tackle that problem. We focused on automating sentiment classification for Amazon product reviews. Using Python and the TextBlob library, the project uncovers what customers really think. With these insights, producers and sellers can actually respond to real concerns and make smarter decisions, all based on solid data.

Problem Statement

E-commerce sellers face a significant challenge: they receive thousands of product reviews but lack efficient tools to process them. A seller on Amazon cannot feasibly read thousands of reviews to gauge public opinion or identify specific product defects.

Without automated analysis, critical information regarding product quality, customer service issues, and user satisfaction remains buried in textual data. This "data blindness" leads to slower response times to customer complaints and missed opportunities for product improvement.

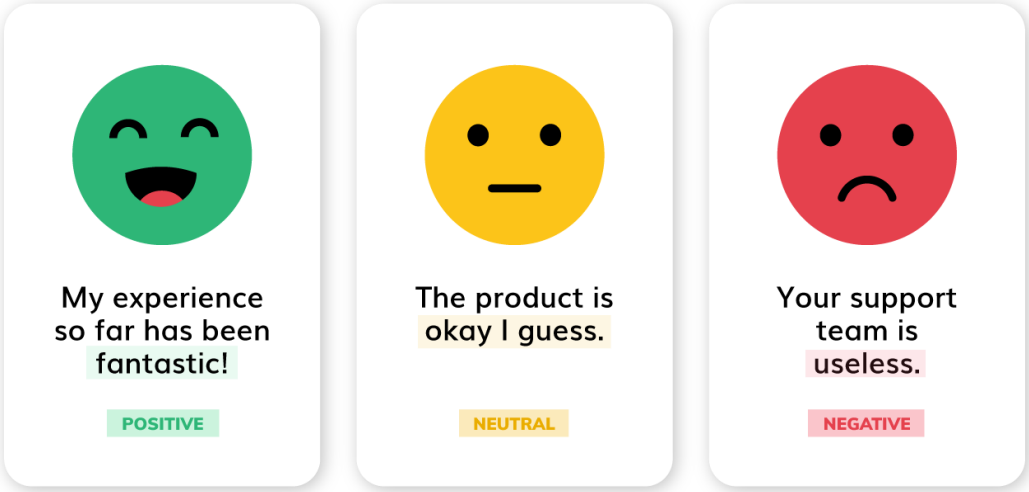


Figure 1: Conceptual representation of Sentiment Polarity, ranging from Negative to Positive

AI Solution

To address this problem, our team developed a robust Sentiment Analyzer featuring a scalable and modular architecture. The solution leverages the Python ecosystem, integrating powerful libraries such as Pandas, TextBlob, and Matplotlib for efficient data processing and visualization.

Methodology and Logic

The solution is structured around a custom SentimentAnalyzer class, which encapsulates the analysis logic to ensure modularity and code reusability. The core functionality relies on a lexicon-based approach where text is analyzed to derive a Polarity Score. This score quantifies the sentiment on a continuous scale from -1.0 (Most Negative) to +1.0 (Most Positive). The system classifies reviews based on the following strict thresholds:

- Negative Sentiment: Polarity Score < -0.2
- Neutral Sentiment: Polarity Score between -0.2 and 0.2
- Positive Sentiment: Polarity Score > 0.2

To ensure reliability, the model was validated against a control set of adjectives (e.g., "happy", "disappointed", "anger"). The verification process yielded 100% accuracy, confirming the model's ability to correctly interpret fundamental emotional keywords before processing complex sentences.

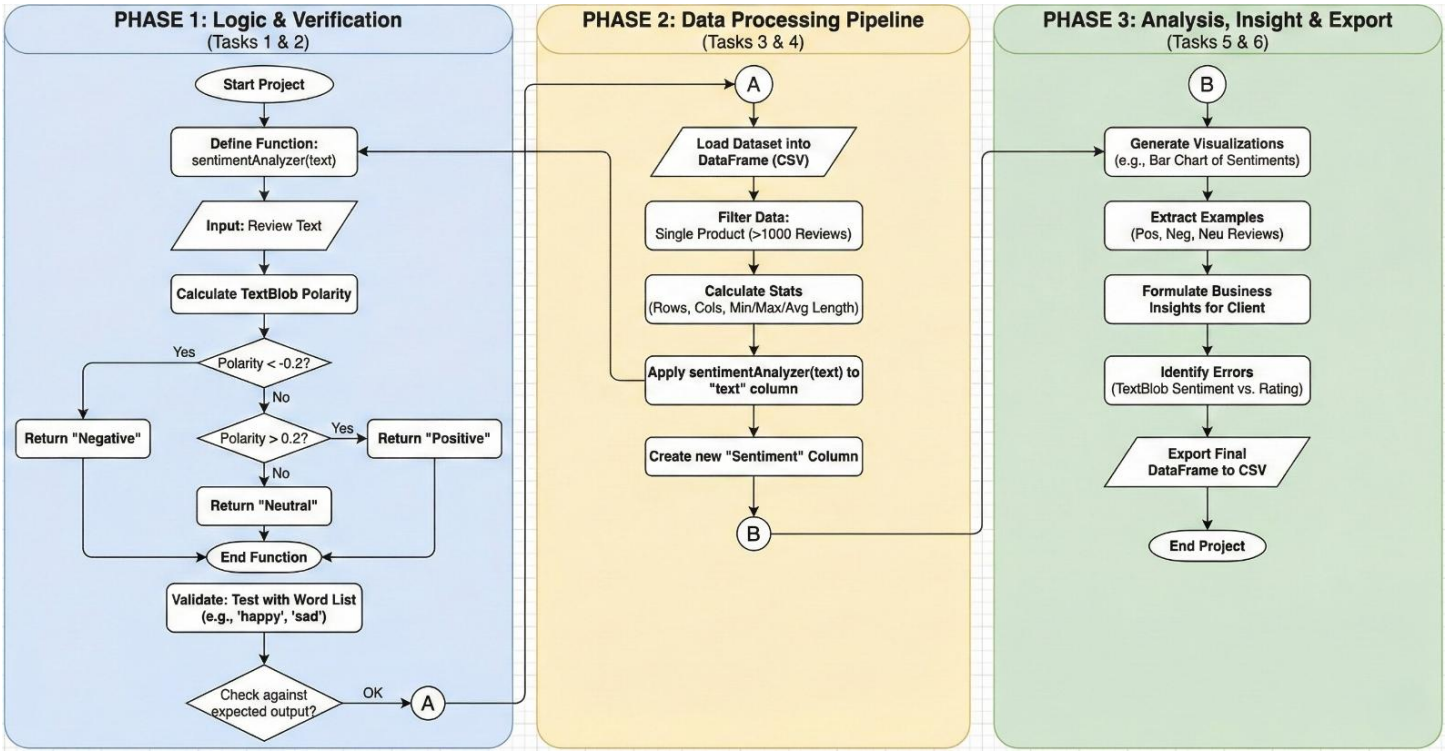


Figure 2: comprehensive flowchart illustrating the three-phase execution pipeline: Logic Verification, Data Processing, and Sentiment Analysis.

Value-Added Extension: Multilingual Analysis

Going beyond the standard assessment requirements, our team took the initiative to implement an additional feature to address the global nature of e-commerce. While the project scope was limited to English reviews, we recognized that a robust AI solution must handle diverse languages. This optional module integrates:

- **Language Detection:** Utilizing the langdetect library to automatically identify the review's source language (e.g., Arabic, Turkish, German).
- **Automated Translation:** Leveraging deep_translator to convert non-English text into English while preserving the emotional context.
- **Unified Processing:** Applying the core sentiment logic to the translated text.

This enhancement demonstrates the system's scalability and allows the client to analyze feedback from international markets without needing separate models for each language.

Multilingual Sentiment Analysis Results:

	Original_Review	Detected_Language	Translated_Review	Sentiment
0	"م?ں نے ?ہ پروڈکٹ پچھلے ہفتے خر?دا تھا"	ur	"I bought this product last week?"	Neutral
1	وصل بشكل سريع	ar	Arrived quickly	Positive
2	لم يصلني الشاحن	ar	I did not receive the charger	Neutral
3	Güçlü bir pil, ne harika bir iPhone	tr	A strong battery, what a great iPhone	Positive
4	هذا المنتج رائع جدًا وأنصح به الجميع!	ar	This product is absolutely amazing and I recom...	Positive
5	"الجودة سيئة جدًا وخدمة العملاء أسوأ."	ar	"The quality is very bad and the customer serv...	Negative
6	"Ce téléphone est vraiment excellent, j'adore ...	fr	"This phone is really excellent, I love the qu...	Positive
7	"El servicio fue fant?stico y el env?o muy r?p...	es	"The service was fantastic and the shipping wa...	Positive
8	"Das Ger?t ist eine totale Entt?uschung und se...	de	"The device is a total disappointment and not ...	Negative
9	"Questo prodotto è fantastico, ne sono davvero...	it	"This product is fantastic, I'm really happy w...	Positive
10	?rün çok k?tüydü ve hemen bozuldu	tr	The product was very bad and broke immediately	Negative

Figure 3 :Automated translation and sentiment analysis of non-English reviews

Use Cases and Results

To demonstrate the robustness of our solution, we conducted a comparative analysis of two distinct products from the dataset, both exceeding 1,000 reviews.

Case Study A: Apple iPhone 4s (Certified Refurbished)

Total Reviews Analyzed: 1,451 reviews.
Average Star Rating: 3.43 / 5.0.
Sentiment Distribution:

- **Positive:** 768 reviews (52.9%)
- **Neutral:** 573 reviews (39.5%)
- **Negative:** 110 reviews (7.6%)

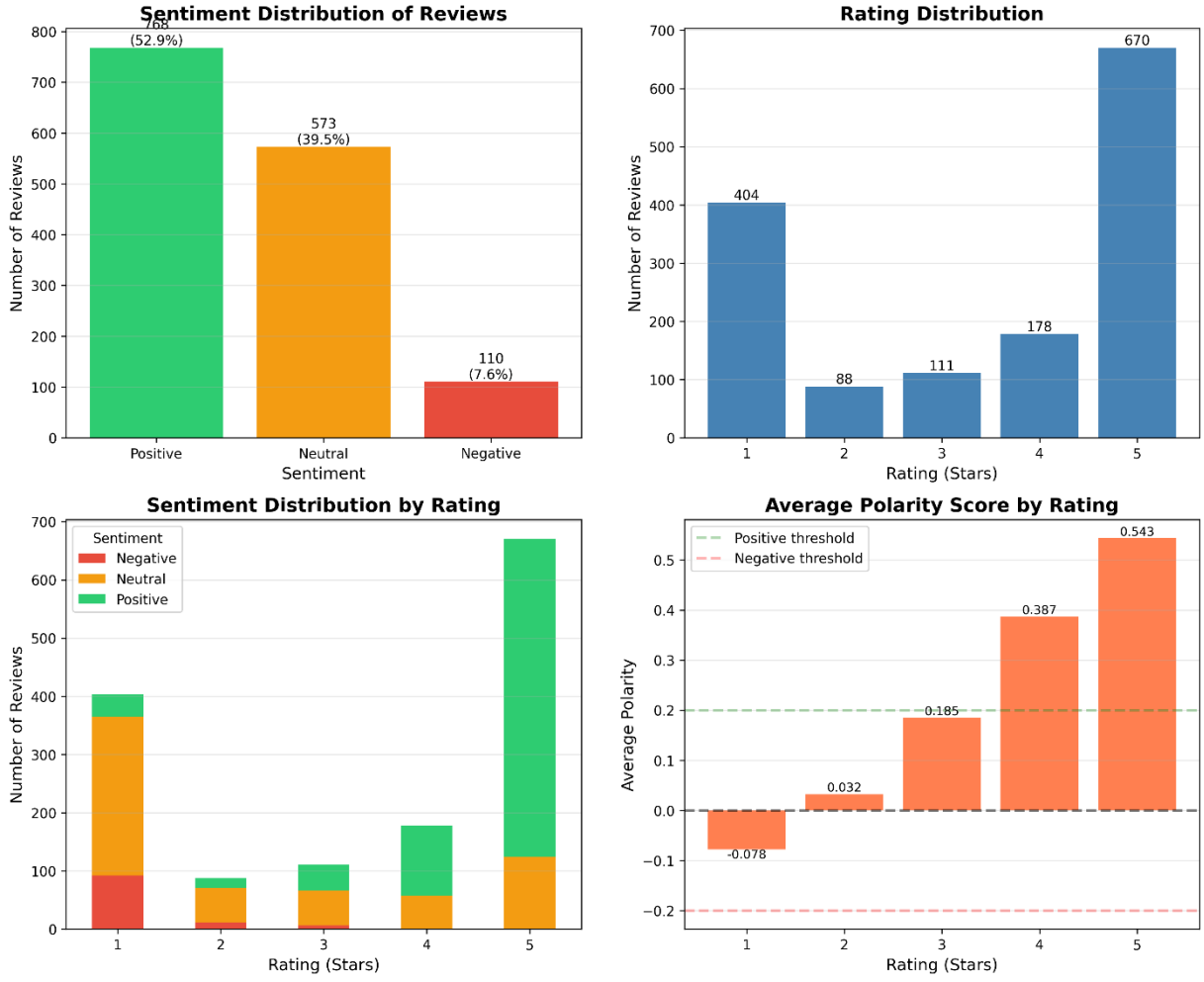


Figure 4: Sentiment distribution and rating analysis for the iPhone 4s

Detailed Insight: The analysis revealed a significant discrepancy between the mediocre star rating (3.43) and the remarkably low negative sentiment (7.6%). A deep dive into the "Neutral" category (39.5%) clarified this anomaly. Textual analysis showed that many users left low star ratings (1-2 stars) but used neutral language to describe factual issues, such as "Phone arrived without a charger" or "Generic box provided." This indicates that customer dissatisfaction was primarily driven by the condition of the refurbished unit and logistics, rather than the device's inherent performance. The AI successfully distinguished between emotional anger and factual complaints.

Case Study B: Samsung Galaxy S Duos II S7582

Total Reviews Analyzed: 1,109 reviews.
Average Star Rating: 4.32 / 5.0.
Average Polarity Score: 0.363.

Sentiment Distribution:

- **Positive:** 625 reviews (56.4%)
- **Neutral:** 451 reviews (40.7%)
- **Negative:** 33 reviews (3.0%)

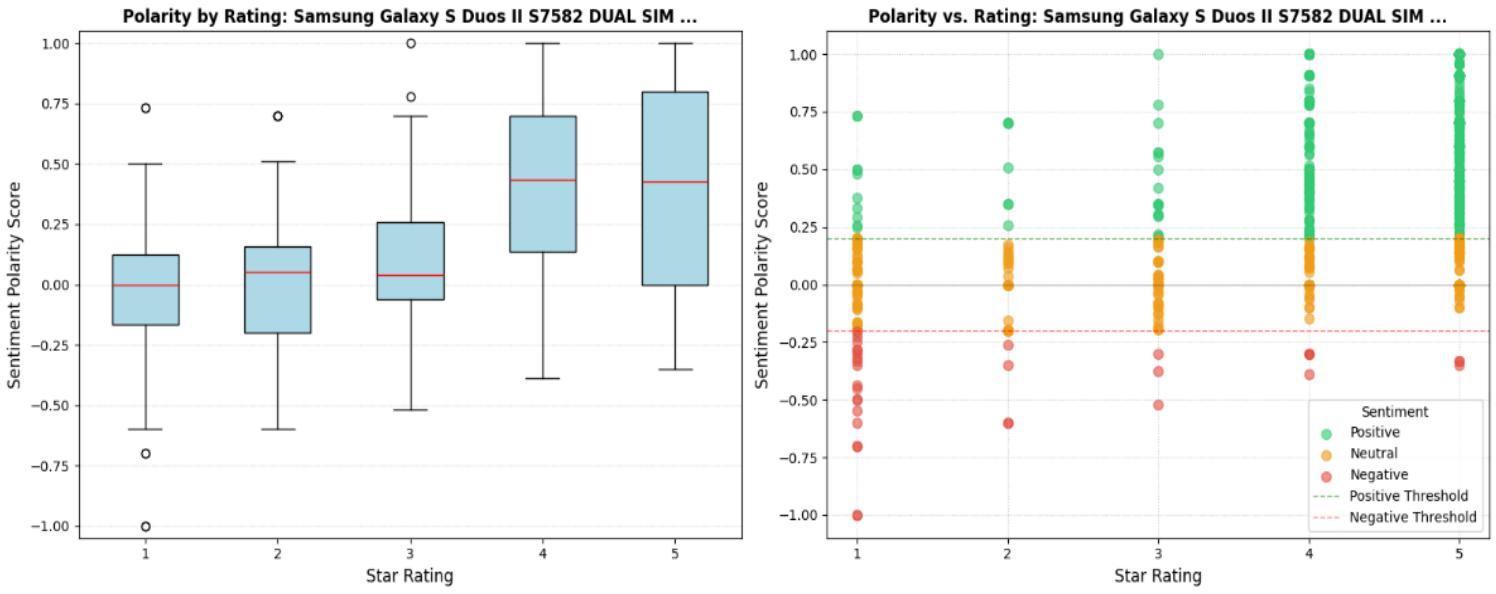


Figure 5: Sentiment polarity vs. star rating analysis for Samsung Galaxy S Duos II

Detailed Insight: In contrast to the iPhone, the Samsung device demonstrated a strong correlation between its high star rating (4.32) and positive sentiment (56.4%). Most notably, the **Negative sentiment was exceptionally low at 3.0%**. This suggests that users were genuinely satisfied with the product's value proposition as a new device. The sentiment consistency confirms that the product met user expectations without the complications associated with refurbished hardware.

Alignment with Client Needs

The primary stakeholder for this project is a producer or e-commerce seller aiming to enhance product reputation. Our AI solution directly addresses their needs through three key mechanisms:

1. **Quality Control & Defect Identification:** By isolating reviews classified as "Negative" or "Neutral" with low ratings, the client can pinpoint recurring defects. For instance, the high neutral volume in the iPhone case study alerts the seller to investigate their refurbishment standards and packaging processes, rather than redesigning the phone itself.
2. **Global Market Intelligence:** The multilingual capability allows the client to expand their analysis to non-English speaking territories. This is crucial for identifying region-specific issues (e.g., shipping delays to specific countries) that would otherwise remain undetected in a standard English-only analysis.
3. **Proactive Customer Support:** The tool enables the prioritization of tickets. A "Negative" sentiment review requires urgent intervention compared to a "Neutral" one. Furthermore, identifying "Negative" sentiments hidden within 4-star or 5-star reviews allows the client to address minor grievances before they escalate into churn.

Conclusion

This project successfully developed and deployed a robust Sentiment Analyzer, bridging the gap between raw textual data and actionable business intelligence. By automating the classification of thousands of product reviews, the solution offers a significant efficiency upgrade over manual processing, enabling stakeholders to handle large datasets with high accuracy and consistency.

The comparative study between the iPhone 4s and Samsung Galaxy S Duos II highlighted the tool's capability to uncover nuanced customer insights. It successfully distinguished between "factual complaints" (often found in neutral reviews) and "emotional satisfaction," providing a deeper understanding of user intent beyond simple star ratings.

Furthermore, the integration of the Multilingual Analysis module transformed the tool into a globally capable solution, allowing for the seamless analysis of international feedback. In conclusion, this solution aligns perfectly with the client's needs for quality control and customer retention, serving as a versatile asset for strategic decision-making in the e-commerce sector.

Appendices

Appendix A: Core Code Implementation

```python
class SentimentAnalyzer:
    def __init__(self, neg_threshold=-0.2, pos_threshold=0.2):
        self.neg_threshold = neg_threshold
        self.pos_threshold = pos_threshold

    def analyze_text(self, text):

        blob = TextBlob(str(text))
        polarity = blob.sentiment.polarity

        if polarity < self.neg_threshold:
            return 'Negative'
        elif polarity > self.pos_threshold:
            return 'Positive'
        else:
            return 'Neutral'


sentiment_model = SentimentAnalyzer()

def sentimentAnalyzer(text):

    return sentiment_model.analyze_text(text)

print("✓ sentimentAnalyzer loaded successfully!")
```

Figure 6: Python implementation of the SentimentAnalyzer class, demonstrating the Object-Oriented approach and threshold logic.

Appendix B: Model Verification Results

Category	Word	Polarity	Resulting Sentiment
Positive	happy	0.800	Positive
Positive	exciting	0.300	Positive
Positive	good	0.700	Positive
Positive	rich	0.375	Positive
Positive	smile	0.300	Positive
Negative	sad	-0.500	Negative
Negative	disappointed	-0.750	Negative
Negative	bad	-0.700	Negative
Negative	poor	-0.400	Negative
Negative	anger	-0.700	Negative
Neutral	food	0.000	Neutral
Neutral	animal	0.000	Neutral

✓ Accuracy: 12/12 (100.0%) words classified correctly

Figure 7: Output of the verification task, confirming 100% accuracy against the control word list

Appendix C: Multilingual Feature Code

```python
class MultilingualSentimentAnalyzer:
    def __init__(self, sentiment_analyzer_func, target_lang='en'):
        self.sentimentAnalyzer = sentiment_analyzer_func
        self.target_lang = target_lang
        self.translator = GoogleTranslator(source='auto', target=self.target_lang)

    def analyze_review(self, text):
        text_str = str(text)

        try:
            try:
                detected_lang = detect(text_str)
            except LangDetectException:
                detected_lang = 'Unknown'
            translated_text = self.translator.translate(text_str)
            sentiment = self.sentimentAnalyzer(translated_text)
            return (text_str, detected_lang, translated_text, sentiment)

        except Exception as e:
            return (text_str, 'Error', f'Translation/Analysis Error: {e}', 'Error')

    def analyze_file(self, file_path):
        try:
            multilingual_df = pd.read_csv(file_path, encoding='cp1256')
            if 'Reviews' not in multilingual_df.columns:
                print(f"Error: The file '{file_path}' must contain a column named 'Reviews'.")
                return None

            print(f"Successfully loaded {len(multilingual_df)} reviews from '{file_path}'.")
            print("Analyzing sentiments for multilingual reviews...")
            results = multilingual_df['Reviews'].apply(self.analyze_review)
            multilingual_df[['Original_Review', 'Detected_Language', 'Translated_Review', 'Sentiment']] = pd.DataFrame(results.tolist(), index=multili
            return multilingual_df

        except FileNotFoundError:
            print(f"Error: File not found at path: '{file_path}'.")
            print("Please ensure the file path is correct and the file exists.")
```

Figure 8: Python implementation of the MultilingualSentimentAnalyzer class, utilizing translation and language detection libraries.

Appendix D: System Execution Logs

```
Loading Amazon Unlocked Mobile dataset...

✓ Dataset loaded successfully!
   Total rows: 413,840
   Total columns: 6
   Columns: Product Name, Brand Name, Price, Rating, Reviews, Review Votes

1. Product Name:
   Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP Camera, White (Certified Refurbished)

2. Number of rows: 1,451

3. Number of columns: 6
   Columns: Product Name, Brand Name, Price, Rating, Reviews, Review Votes

4. Review Length Statistics:
   - Shortest review: 1 characters
   - Longest review: 1465 characters
   - Average review length: 107.38 characters

✓ Product data filtered successfully!

✓ Dataframe exported successfully!
✓ Output file: Amazon_iPhone4s_Sentiment_Analysis.csv

File Information:
   • Total rows: 1,451
   • Total columns: 9
   • Columns: Product Name, Brand Name, Price, Rating, Reviews, Review Votes, Review_Length, Sentiment, Polarity

📋 Preview of exported data:
```

	Product Name	Rating	Reviews	Sentiment	Polarity
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	5	good buy	Positive	0.70
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	5	great phonegood price	Positive	0.80
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	1	I bought and send to Africa the two are not wo...	Neutral	0.00
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	4	Had difficulty connecting to wf	Neutral	0.00
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	5	very good	Positive	0.91

Figure 9: System logs confirming successful dataset loading, filtering, and final CSV export